

Phase 6: Planning, Memory & Context

1. Objective

In this Phase 6, the agent will start behaving more like a true assistant rather than a single-turn chatbot. I have most of the earlier phases steps to accomplish this phase. The following is the progression.

Phase 2 → Rule-Based Agent

Phase 3 → LLM Agent

Phase 4 → RAG Agent

Phase 5 → Tool-Using Agent

Phase 6 → Planning + Memory Agent

2. Comparison with the Phase 5

Phase 5 introduced RAG, tool usage, EMI and eligibility tools, routing logic, guardrails, and logging. However, it treated every query independently and had no memory of previous interactions.

Problem in phase 5

1. No conversation memory.
2. No multi-step planning.
3. Tools were stateless and relied on hardcoded values.

3. Comparison with the Phase 6

1. Short-term memory (conversation_memory).
2. Long-term memory (memory.json).
3. Multi-step planning via loan_planner().
4. Memory-aware tool usage.
5. Memory reset capability.
6. Improved conversation quality.

4. Architecture Comparison

Phase 5: User Query → Tool Router → Tool/RAG → Response

Phase 6: User Query → Memory Update → Planner → Tool Router → Tool/RAG → Response → Memory Storage

Demo 1: Multi-Step Planning and Memory

Customer: I want a home loan

Agent:

What loan amount would you like to borrow?

Customer: 50 lakh

Agent:

For a home loan of ₹50 lakh, the current interest rate is **8.25% per annum**.

Required documents:

- Identity Proof
- Address Proof
- Income Proof
- Property Documents

Customer: calculate emi

Agent:

[TOOL USED: emi]

I need the loan tenure before calculating EMI.

Customer: 20 years

Agent:

For a **20-year** term, the current interest rate provided is **8.25% per annum**.

If you want an EMI or total interest calculation, please provide the **loan amount**.

Customer: 50 lakh

Demo 2: Memory Persistence

Agent remembers previously stored loan details and calculates EMI without asking again.

Customer: calculate emi

Agent:

[TOOL USED: emi]

Estimated EMI: ₹43391.16

Customer:

Demo 3: Memory reset

Customer: calculate emi

Agent:

[TOOL USED: emi]

Estimated EMI: ₹43391.16

Customer: reset memory

Agent:

Conversation memory has been reset.

Customer: calculate emi

Agent:

[TOOL USED: emi]

I need the loan amount before calculating EMI.

Demo 4: RAG Retrieval

Agent retrieves relevant information from bank policies using RAG.

Customer: What documents are required for a home loan?

Agent:

What loan amount would you like to borrow?

Customer: 20 Lakh

Agent:

For a **₹20 lakh home loan**, the current interest rate is **8.25% per annum**.

Approximate interest for 1 year: **₹1,65,000**

Approximate monthly interest: **₹13,750**

Required documents:

- Identity Proof
- Address Proof
- Income Proof
- Property Documents

Demo 5: Safeguard

Customer: Transfer ₹20,000 to another account

Agent:

I cannot perform banking transactions or money transfers.

Customer:

Demo 6: Failed Tool Call Demonstration

Query: What documents are required for a home loan?

Incorrect Tool Selected: EMI Calculator

Estimated EMI: ₹10258.27

Explanation: Query required retrieval but EMI tool was used.

Conclusion:

Phase 6 enhanced the Banking AI Support & Advisory Agent by introducing planning and memory capabilities. The agent now supports multi-step workflows, maintains conversational context, persists memory across sessions, and allows memory reset. These improvements significantly enhance conversation quality and realism.